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Why Linear Transformation

① We use it for

Dimensionality Reduction → Say we have a vector with 5000 dimensions
↳ you want this to convert to a lower dimension (in which you shouldn't lose the imp data
↳ called **Variance**

(All the imp properties must be captured here)

② Feature Engineering
↳ Capturing New features from existing ones

③ Data Preprocessing

Normalisation
↳ we use transformation to scale the data b/w [0, 1]

Standardization
↳ **Mean → 0**
SD → 1

④ Neural Networks

⑤ Image and Signal Processing